

1(a)	$m = \rho V$ or ρAL	C1
	$W = mg$	C1
	$(A =) 24 / (9.81 \times 850 \times 0.18) = 0.016 \text{ (m}^2\text{)}$	A1
	or	
	$P = F / A$	(C1)
	$P = \rho gh$	(C1)
	$(A =) 24 / (9.81 \times 850 \times 0.18) = 0.016 \text{ (m}^2\text{)}$	(A1)
1(b)(i)	$(\text{upthrust} =) 24 + 8(.0) = 32 \text{ (N)}$	A1
1(b)(ii)	$(\Delta)p = 32 / 0.016 (= 2000)$	C1
	$(\Delta)p = \rho g(\Delta)h$	C1
	$\rho = 2000 / (9.81 \times 0.17)$	
	$= 1200 \text{ kg m}^{-3}$	A1
1(c)(i)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \text{area under graph}$	C1
	$(\Delta)E = (\frac{1}{2} \times 8.0 \times 0.40) - (\frac{1}{2} \times 4.0 \times 0.20)$ or $(\frac{1}{2} \times 20 \times 0.40^2) - (\frac{1}{2} \times 20 \times 0.20^2)$ or $\frac{1}{2} \times (4.0 + 8.0) \times 0.20$	C1
	$= 1.2 \text{ J}$	A1
1(c)(ii)	length = 30 cm	A1